ModelIt!

Student Activity Pack

Can We Engineer the Amazon Back to Life?

Modeling Rainforest Restoration and Ecological Tipping Points

Grade 10L2 | NGSS HS-LS2-7, HS-ESS2-7 | Stability and Change

STUDENT NAME _____ DATE _____

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think happens to the land, soil, and climate after a rainforest is cut down?

2 Why might it be harder to regrow a rainforest than to plant a new forest in a temperate region?

3 Draw a diagram showing how you think a rainforest creates its own weather system.

4 What do you think a 'tipping point' means for an ecosystem like the Amazon?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Deforestation Rate | Reforestation Effort | Soil Nutrient Recovery | Water Cycle Restoration | Biodiversity Return | Carbon Sequestration | Tipping Point Threshold

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Continued Deforestation

Current deforestation rate, minimal restoration

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Aggressive Restoration

Zero deforestation, maximum replanting

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Partial Intervention

Halved deforestation, moderate restoration

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Fire Vulnerability — The susceptibility of degraded and edge-of-forest areas to fire during dry seasons, which can destroy restoration progress and accelerate tipping point approach

Indigenous Land Management — The conservation effectiveness of indigenous-managed territories, which show 50% lower deforestation rates than other areas due to traditional ecological knowledge and stewardship

Economic Pressure — The market forces driving deforestation — cattle ranching, soy farming, illegal logging, and gold mining — which must be addressed for any restoration plan to succeed long-term

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

The Amazon rainforest is approaching a tipping point where it could transform from the world's largest carbon sink into a carbon source — can engineering and restoration reverse the damage?
Use evidence from your simulation to support your answers.

1 How does your model demonstrate that the Amazon's water cycle is a self-sustaining feedback loop — and why its disruption is so dangerous?

2 What does your model reveal about the timescale mismatch between the speed of destruction and the pace of restoration?

3 Why is the tipping point concept so critical for policy decisions about the Amazon?

4 How does Biodiversity Return differ from simply replanting trees, and why does this distinction matter for ecosystem function?

5 What are the limitations of modeling complex tropical ecology with only seven components when real rainforests contain millions of interacting species?

8 STEM Challenge: Design an Amazon Restoration Master Plan

ModelIt!

YOUR ENGINEERING MISSION

Design a 50-year restoration plan for a 10,000 square kilometer degraded Amazon region, specifying planting strategies, soil recovery techniques, and monitoring milestones.

THE SCENARIO

The Brazilian government has committed to restoring 10,000 square kilometers of degraded Amazon land and hired your team to design the master plan. You must balance speed (preventing tipping point crossing) with ecological integrity (ensuring the restored forest is self-sustaining, not just a tree plantation).

Engineering Design Thinking

1. WHAT PLANTING DENSITY, SPECIES MIX, AND SPATIAL ARRANGEMENT WOULD MAXIMIZE EARLY SOIL RECOVERY?
2. HOW WOULD YOU USE CORRIDORS TO CONNECT RESTORED PATCHES WITH INTACT FOREST FOR WILDLIFE COLONIZATION?
3. WHAT ROLE WOULD LOCAL AND INDIGENOUS COMMUNITIES PLAY IN LONG-TERM FOREST STEWARDSHIP?
4. WHAT METRICS WOULD YOU USE TO DETERMINE SUCCESS AT 5, 10, 25, AND 50 YEAR MILESTONES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Restoration Ecologists design and implement ecosystem recovery projects for degraded landscapes. They work for conservation organizations, government agencies, and consulting firms, earning $55,000–$110,000/year. Tropical Ecologists studying Amazon dynamics earn $60,000–$130,000/year.